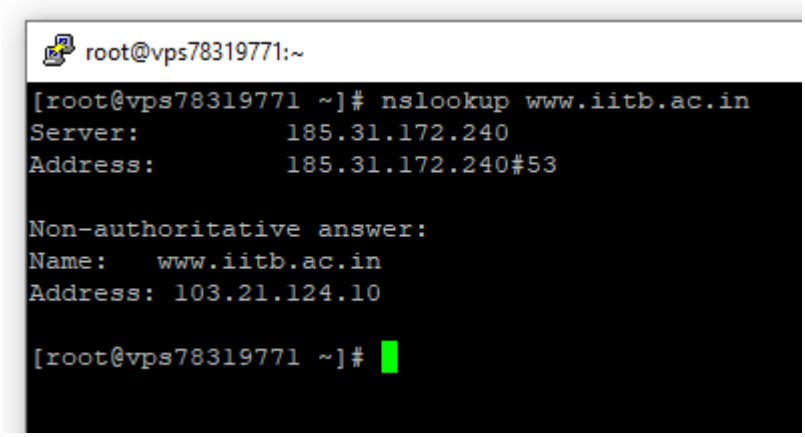
# CSE4074 Computer Networks
# Homework II

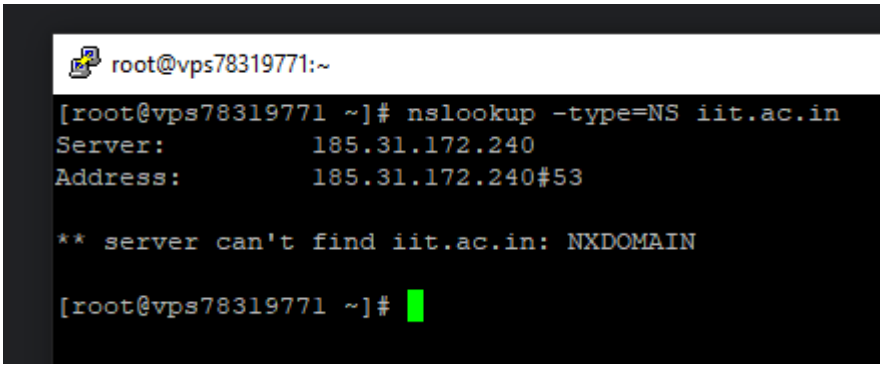
I used both Windows and Linux operating systems to complete this homework due to the limited capabilities of Windows' command prompt. Because of that the screenshots may look different for each section of the homework.

## nslookup

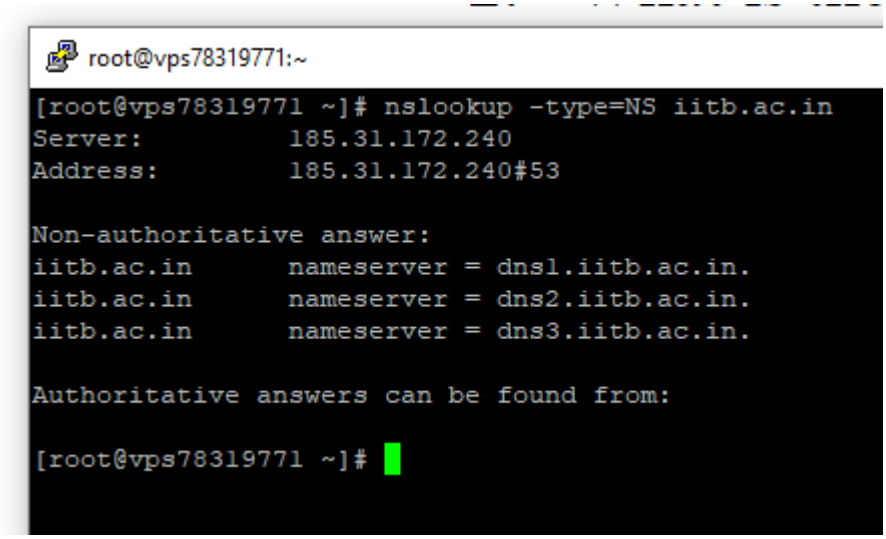
1. 103.21.124.10



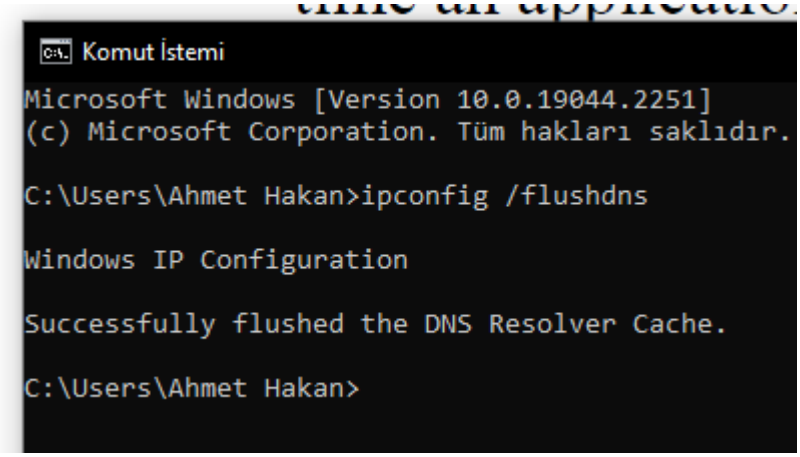

2. 185.31.712.240
3. It came from non-authoritative server.
4. This domain is not available anymore.



But NS lookup for iitb.ac.in is given below.

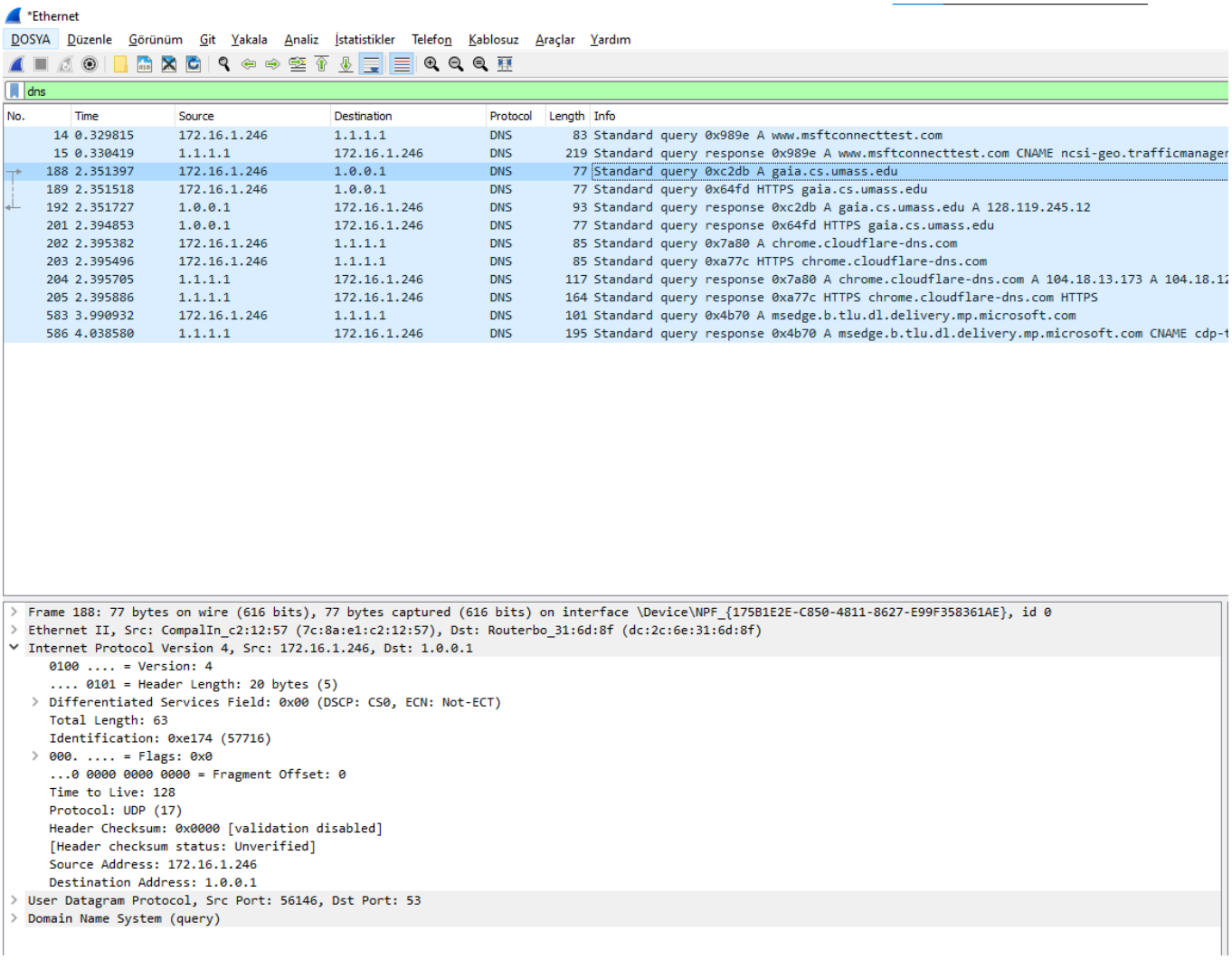



The DNS cache on your computer

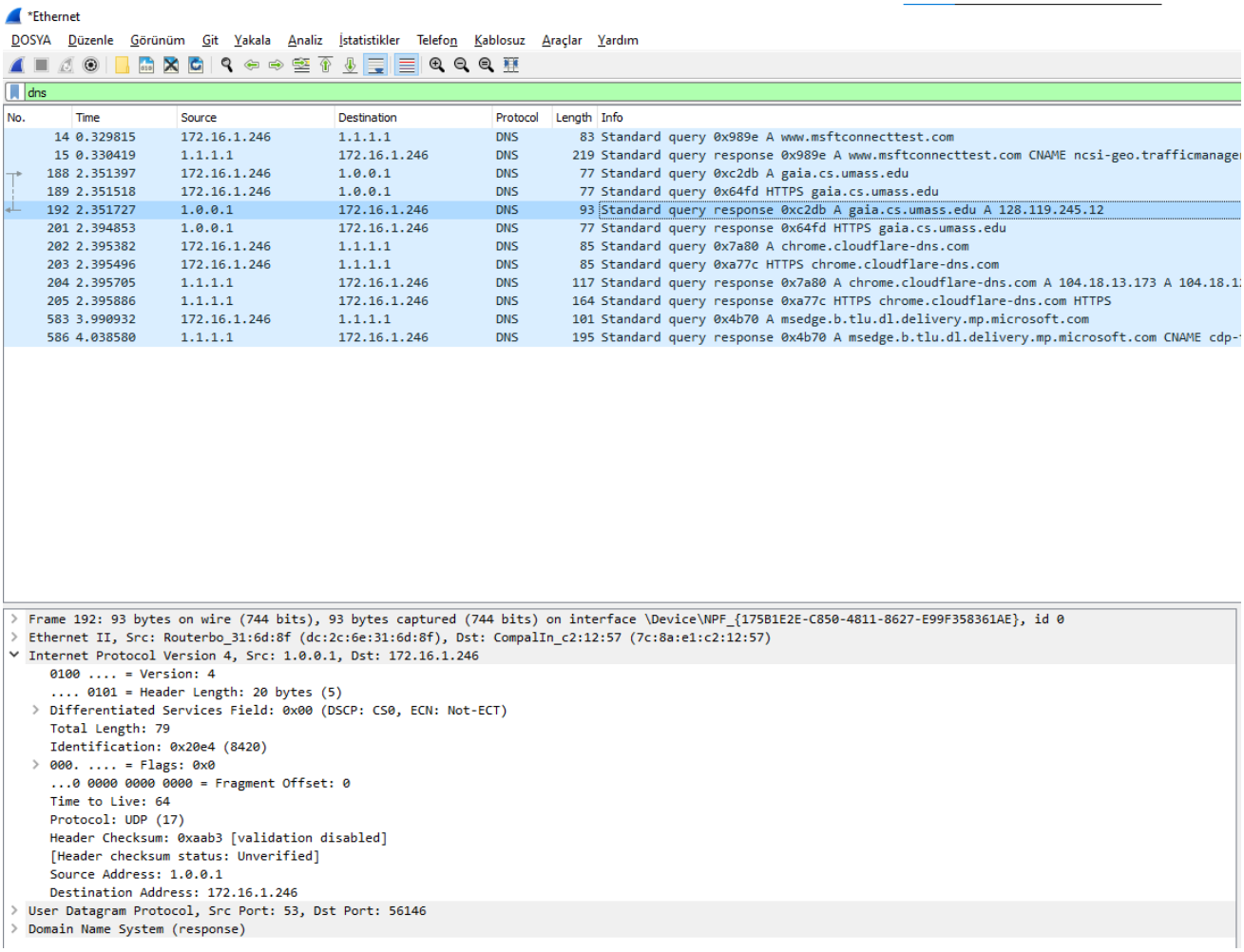



Tracing DNS with Wireshark

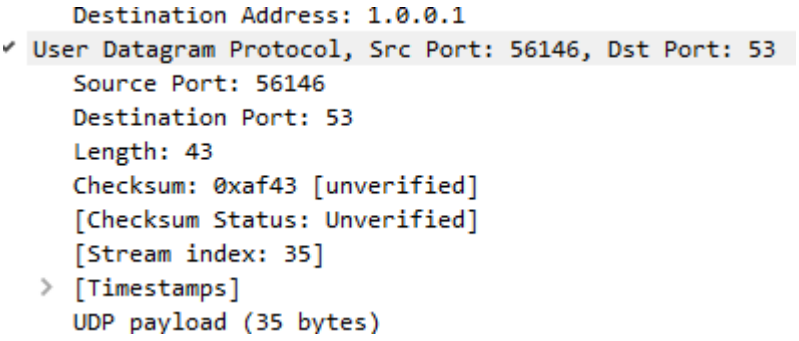
5. The packet number is 188. And it is sent over UDP protocol.



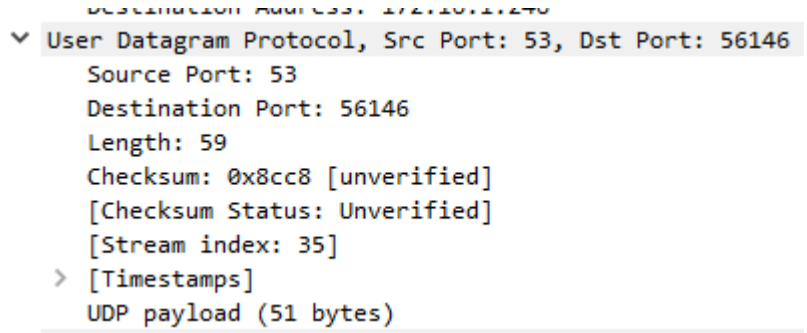
6. The packet number is 192 and the response is received over UDP.



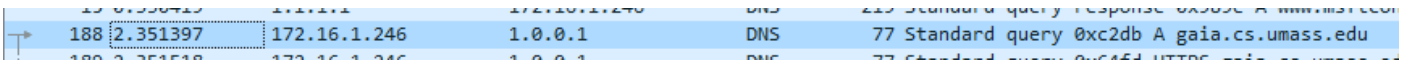

7. Destination port of DNS query message is 53




Source port of DNS response message is 53

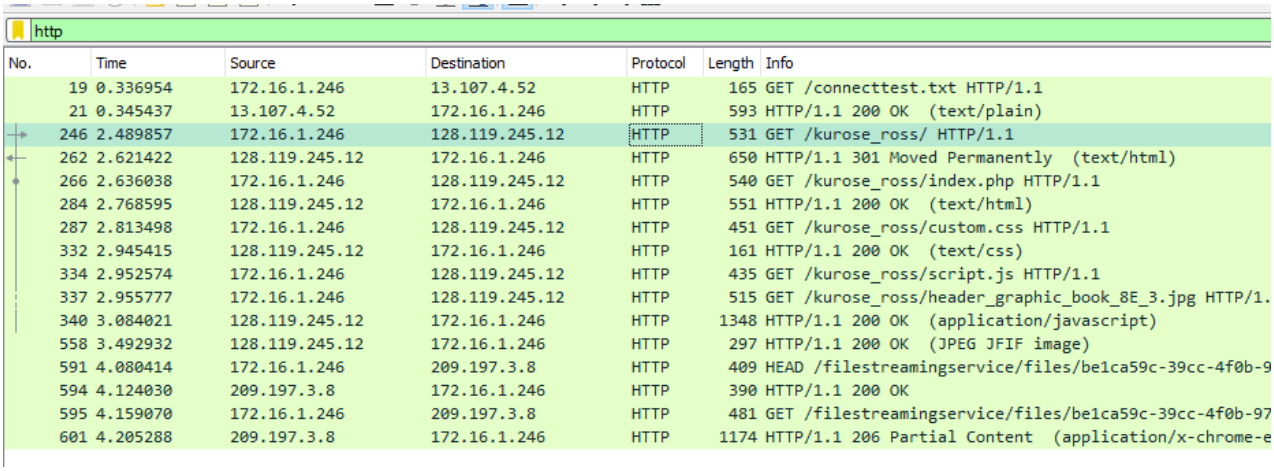



8. DNS query is sent to 1.0.0.1



9. It has one question. It asks the IP address of gaia.cs.umass.edu.
   It does not have any answers.
10. It does not ask for anything. It has an answer for the previous question, so it returns the IP address of gaia.cs.umass.edu
11.

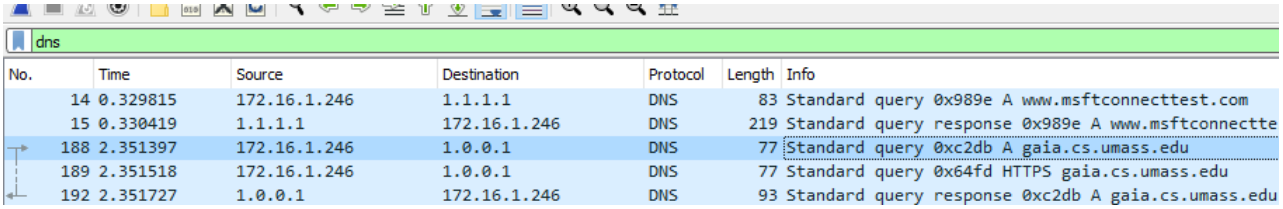
11.1. The packet number is 246



http

| No. | Time | Source | Destination | Protocol | Length | Info |
|---|---|---|---|---|---|---|
| 19 | 0.336954 | 172.16.1.246 | 13.107.4.52 | HTTP | 165 | GET /connecttest.txt HTTP/1.1 |
| 21 | 0.345437 | 13.107.4.52 | 172.16.1.246 | HTTP | 593 | HTTP/1.1 200 OK (text/plain) |
| 246 | 2.489857 | 172.16.1.246 | 128.119.245.12 | HTTP | 531 | GET /kurose_ross/ HTTP/1.1 |
| 262 | 2.621422 | 128.119.245.12 | 172.16.1.246 | HTTP | 650 | HTTP/1.1 301 Moved Permanently (text/html) |
| 266 | 2.636038 | 172.16.1.246 | 128.119.245.12 | HTTP | 540 | GET /kurose_ross/index.php HTTP/1.1 |
| 284 | 2.768595 | 128.119.245.12 | 172.16.1.246 | HTTP | 551 | HTTP/1.1 200 OK (text/html) |
| 287 | 2.813498 | 172.16.1.246 | 128.119.245.12 | HTTP | 451 | GET /kurose_ross/custom.css HTTP/1.1 |
| 332 | 2.945415 | 128.119.245.12 | 172.16.1.246 | HTTP | 161 | HTTP/1.1 200 OK (text/css) |
| 334 | 2.952574 | 172.16.1.246 | 128.119.245.12 | HTTP | 435 | GET /kurose_ross/script.js HTTP/1.1 |
| 337 | 2.955777 | 172.16.1.246 | 128.119.245.12 | HTTP | 515 | GET /kurose_ross/header_graphic_book_8E_3.jpg HTTP/1. |
| 340 | 3.084021 | 128.119.245.12 | 172.16.1.246 | HTTP | 1348 | HTTP/1.1 200 OK (application/javascript) |
| 558 | 3.492932 | 128.119.245.12 | 172.16.1.246 | HTTP | 297 | HTTP/1.1 200 OK (JPEG JFIF image) |
| 591 | 4.080414 | 172.16.1.246 | 209.197.3.8 | HTTP | 409 | HEAD /filestreamingservice/files/be1ca59c-39cc-4f0b-9 |
| 594 | 4.124030 | 209.197.3.8 | 172.16.1.246 | HTTP | 390 | HTTP/1.1 200 OK |
| 595 | 4.159070 | 172.16.1.246 | 209.197.3.8 | HTTP | 481 | GET /filestreamingservice/files/be1ca59c-39cc-4f0b-97 |
| 601 | 4.205288 | 209.197.3.8 | 172.16.1.246 | HTTP | 1174 | HTTP/1.1 206 Partial Content (application/x-chrome-e |

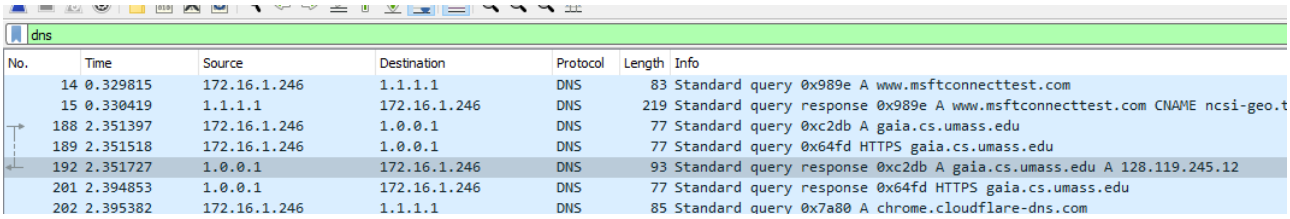
11.2. 188



dns

| No. | Time | Source | Destination | Protocol | Length | Info |
|---|---|---|---|---|---|---|
| 14 | 0.329815 | 172.16.1.246 | 1.1.1.1 | DNS | 83 | Standard query 0x989e A www.msftconnecttest.com |
| 15 | 0.330419 | 1.1.1.1 | 172.16.1.246 | DNS | 219 | Standard query response 0x989e A www.msftconnectte |
| 188 | 2.351397 | 172.16.1.246 | 1.0.0.1 | DNS | 77 | Standard query 0xc2db A gaia.cs.umass.edu |
| 189 | 2.351518 | 172.16.1.246 | 1.0.0.1 | DNS | 77 | Standard query 0x64fd HTTPS gaia.cs.umass.edu |
| 192 | 2.351727 | 1.0.0.1 | 172.16.1.246 | DNS | 93 | Standard query response 0xc2db A gaia.cs.umass.edu |

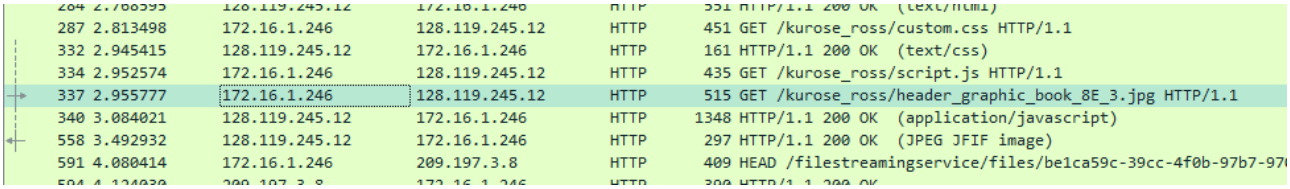
11.3. 192



dns

| No. | Time | Source | Destination | Protocol | Length | Info |
|---|---|---|---|---|---|---|
| 14 | 0.329815 | 172.16.1.246 | 1.1.1.1 | DNS | 83 | Standard query 0x989e A www.msftconnecttest.com |
| 15 | 0.330419 | 1.1.1.1 | 172.16.1.246 | DNS | 219 | Standard query response 0x989e A www.msftconnecttest.com CNAME ncsi-geo.t |
| 188 | 2.351397 | 172.16.1.246 | 1.0.0.1 | DNS | 77 | Standard query 0xc2db A gaia.cs.umass.edu |
| 189 | 2.351518 | 172.16.1.246 | 1.0.0.1 | DNS | 77 | Standard query 0x64fd HTTPS gaia.cs.umass.edu |
| 192 | 2.351727 | 1.0.0.1 | 172.16.1.246 | DNS | 93 | Standard query response 0xc2db A gaia.cs.umass.edu A 128.119.245.12 |
| 201 | 2.394853 | 1.0.0.1 | 172.16.1.246 | DNS | 77 | Standard query response 0x64fd HTTPS gaia.cs.umass.edu |
| 202 | 2.395382 | 172.16.1.246 | 1.1.1.1 | DNS | 85 | Standard query 0x7a80 A chrome.cloudflare-dns.com |

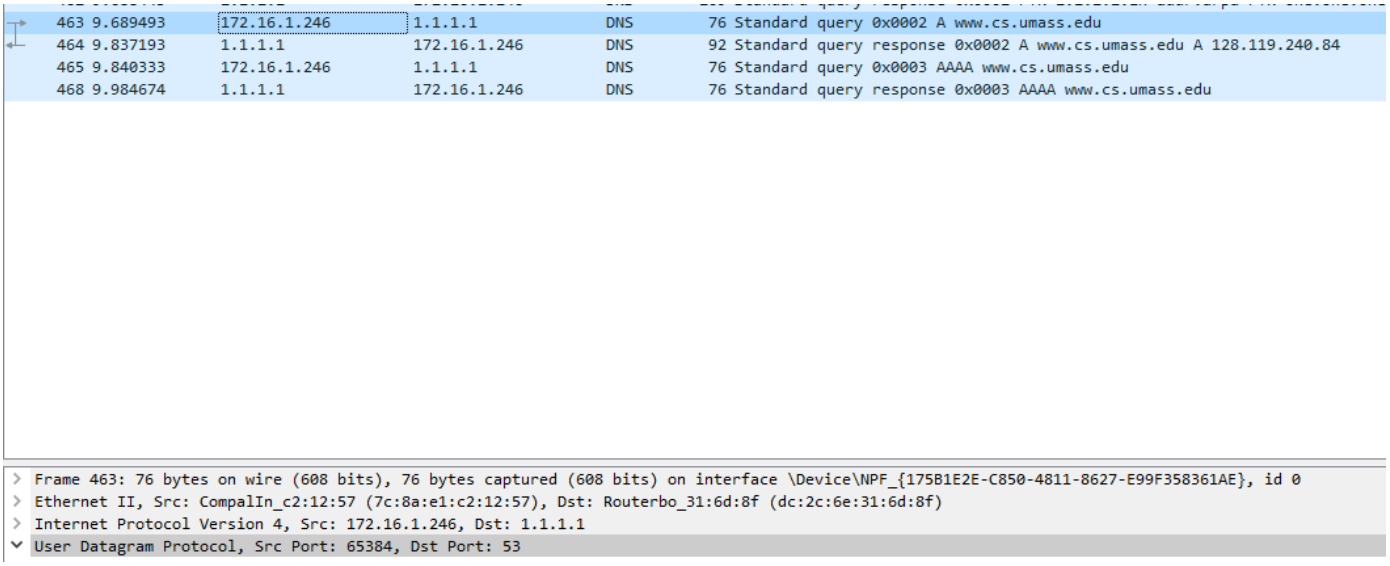
11.4. 337



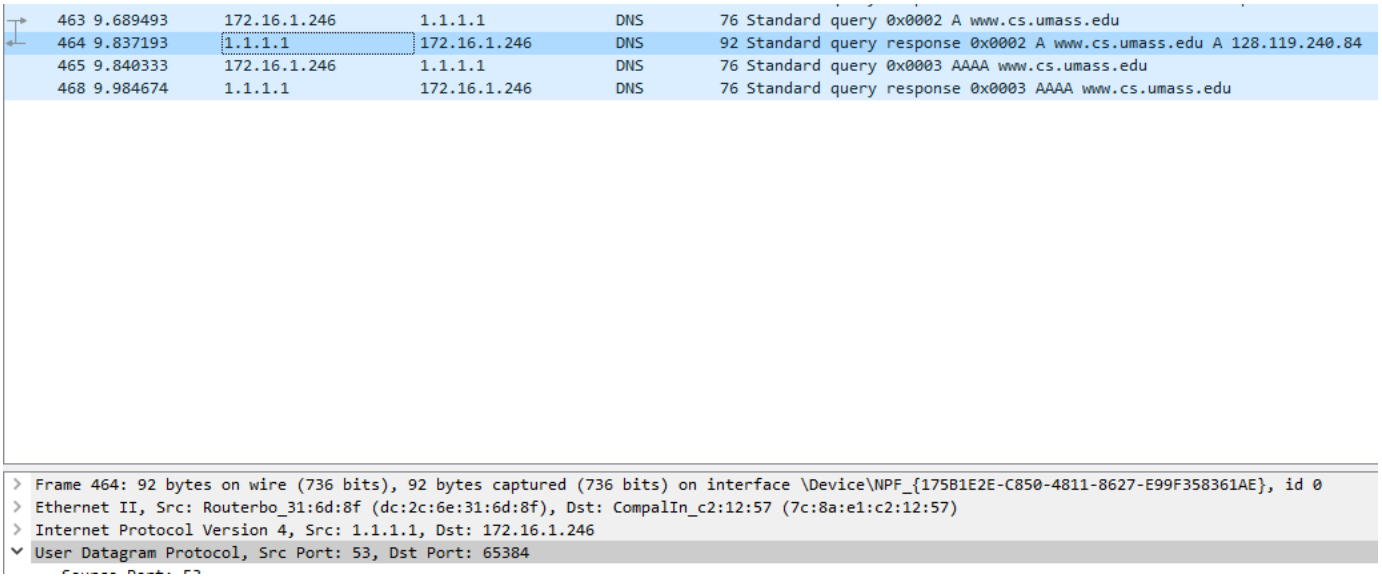
| | | | | | | |
|---|---|---|---|---|---|---|
| 287 | 2.813498 | 172.16.1.246 | 128.119.245.12 | HTTP | 451 | GET /kurose_ross/custom.css HTTP/1.1 |
| 332 | 2.945415 | 128.119.245.12 | 172.16.1.246 | HTTP | 161 | HTTP/1.1 200 OK (text/css) |
| 334 | 2.952574 | 172.16.1.246 | 128.119.245.12 | HTTP | 435 | GET /kurose_ross/script.js HTTP/1.1 |
| 337 | 2.955777 | 172.16.1.246 | 128.119.245.12 | HTTP | 515 | GET /kurose_ross/header_graphic_book_8E_3.jpg HTTP/1.1 |
| 340 | 3.084021 | 128.119.245.12 | 172.16.1.246 | HTTP | 1348 | HTTP/1.1 200 OK (application/javascript) |
| 558 | 3.492932 | 128.119.245.12 | 172.16.1.246 | HTTP | 297 | HTTP/1.1 200 OK (JPEG JFIF image) |
| 591 | 4.080414 | 172.16.1.246 | 209.197.3.8 | HTTP | 409 | HEAD /filestreamingservice/files/be1ca59c-39cc-4f0b-97b7-97 |

11.5. There is no second DNS request because the first request is cached.

12. DNS query destination port is 53




DNS response source port is 53

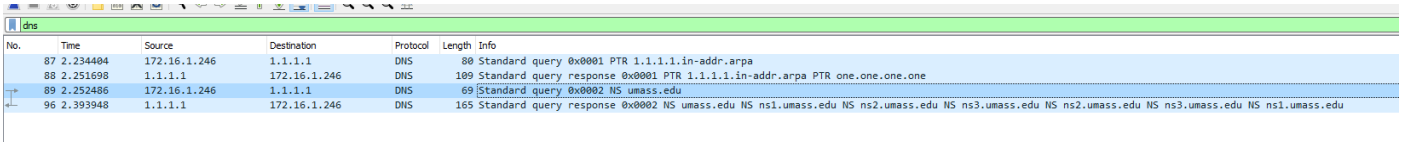



13. DNS request is sent to 1.1.1.1 as can be seen in the screenshot above.
I use a custom DNS server it is not my local DNS server.

14. It asks for A record of the domain. A stands for address of the domain. It does not contain any answers.

15. It does not have any questions it returns an answer for the A record of www.cs.umass.edu according to DNS query made previously

16. The request is sent to 1.1.1.1. It is not my local DNS server. It is a free DNS server provided by Cloudflare. I use it on my computer.



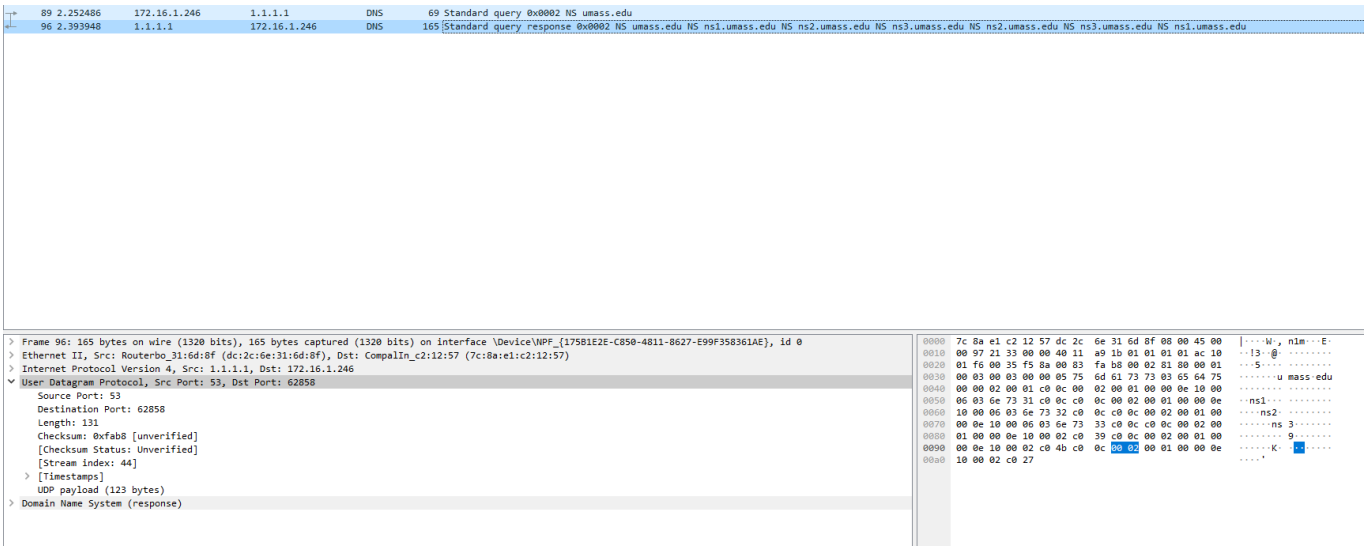

17. It has just one question. It asks for the NS record of umass.edu.

18. It has one answer. It returns NS record of umass.edu but it contains different domains as can be seen below.

```
C:\Users\Ahmet Hakan>nslookup -type=NS umass.edu
Server:  one.one.one.one
Address:  1.1.1.1

Non-authoritative answer:
umass.edu       nameserver = ns1.umass.edu
umass.edu       nameserver = ns2.umass.edu
umass.edu       nameserver = ns3.umass.edu

umass.edu       nameserver = ns2.umass.edu
umass.edu       nameserver = ns3.umass.edu
umass.edu       nameserver = ns1.umass.edu
```




It returned the same nameservers in a different order.

It totally returned three unique nameserver for umass.edu.